

17. A fellow student found in the performance data for his new car the velocity-versus-time graph shown in Fig. 2.

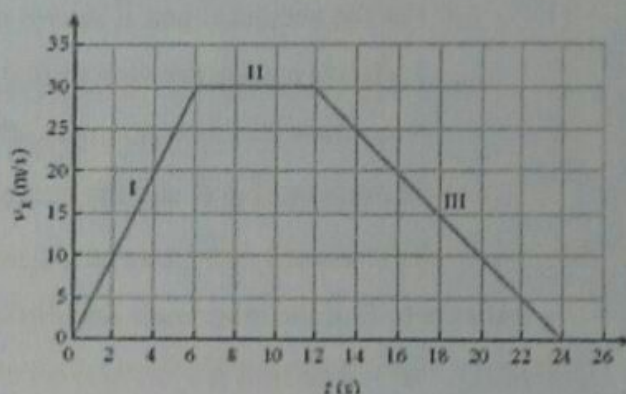


Fig. 2

- a) Find the average acceleration of the car during each of the segments I, II, and III.
b) What is the distance traveled by the car from $t = 0$ s to $t = 6$ s? from $t = 6$ s to $t = 12$ s? from $t = 12$ s to $t = 24$ s?

18. You are on the roof of the physics building, 46.0 m above the ground (Fig. 3). Your friend, who is 1.80 m tall, is walking alongside the building at a constant speed of 1.20 m/s. If you wish to drop an egg on your friend's head, where should he be when you release the egg?

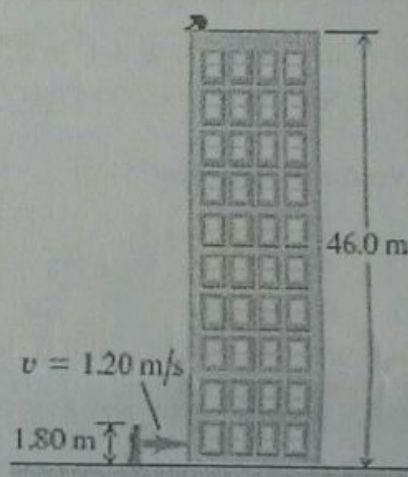


Fig. 3

Assume that the egg is in free fall.

19. Starting from a pillar, you run 200 m east (the $+x$ -direction) at an average speed of 5.0 m/s, and then run 280 m west at an average speed of 4.0 m/s to a post. Calculate (a) your average speed from pillar to post and (b) your average velocity from pillar to post.
20. A car is stopped at a traffic light. It then travels along a straight road so that its distance from the light is given by $x(t) = bt^2 - ct^3$, where $b = 2.40 \text{ m/s}^2$ and $c = 0.120 \text{ m/s}^3$. (a) Calculate the average velocity of the car for the time interval $t = 0$ to $t = 10.0$ s. (b) Calculate the instantaneous velocity of the car at $t = 0$, $t = 5.0$ s, and $t = 10.0$ s. (c) How long after starting from rest is the car again at rest?

13. For the vectors $\vec{A}$ and $\vec{B}$ shown in Fig.

1, (I) use a scale drawing to find the magnitude and direction of (a) the vector sum $\vec{A} + \vec{B}$ and (b) the vector difference $\vec{A} - \vec{B}$. Use your

answers to find the magnitude and direction of (c) $-\vec{A} - \vec{B}$ and (d) $\vec{B} - \vec{A}$.

(II) Find the x and y components of each vector and calculate (a) the magnitude and direction of $2\vec{A} - 3\vec{B} + \vec{D}$, $\vec{A} + \vec{B} + \vec{C} + \vec{D}$, and $-2\vec{C} - 5\vec{B} + 4\vec{D}$;

(b) find a unit vector in the direction of

$\vec{A} + \vec{B}$, $\vec{B} - \vec{A}$, and $\vec{A} + \vec{B} + \vec{C} + \vec{D}$.

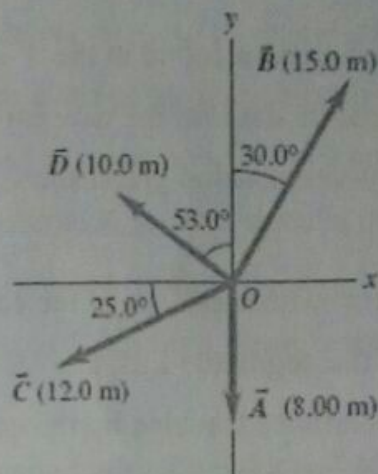


Fig. 1

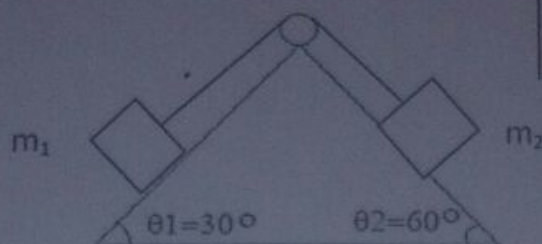
14. A spelunker is surveying a cave. She follows a passage 180 m straight west, then 210 m in a direction 45° east of south, and then 280 m at 30° east of north. After a fourth unmeasured displacement, she finds herself back where she started. Use a scale drawing to determine the magnitude and direction of the fourth displacement.

15. A car travels north at 30.0 m/s for 10.0 min. It then travels south at 40.0 m/s for 20.0 min. What are the total distance the car travels and its displacement?

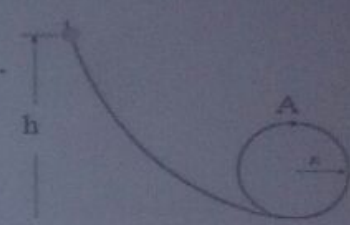
16. You ride your bike along a straight line from your house to a store 1000 m away. On your way back, you stop at a friend's house which is halfway between your house and the store.

- What is your displacement?
- What is the total distance traveled? After talking to your friend, you continue to your house. When you arrive back at your house,
- What is your displacement?
- What is the distance you have traveled?

6. Mass m_1 and m_2 are located on inclined planes of angles 30° and 60° respectively. They are connected by an inextensible string of negligible mass which passes over a smooth peg at A. The coefficient of friction between the masses and the incline is μ . (See the fig.)
- Find the acceleration for any angle θ_1 and θ_2 .
 - Using the angles value, determine the direction of motion if $\mu = 0$ and $m_1 = m_2$.



$\sin 30^\circ = \cos 60^\circ = 0.50$ $\cos 30^\circ = \sin 60^\circ = 0.866$

7. A bead slides without friction around a loop (see fig). The bead is released from a height $h = 3.5R$.
- What is its speed at point A?
 - How large is the normal force on it if its mass is 5.0 gm ?
- 
8. When a 4 kg object is hung vertically on a certain light spring that obeys Hooke's law, the spring stretches 2.50 cm . If the 4 kg object is removed, (a) how far will the spring stretch if a 1.50-kg block is hung on it and (b) how much work must an external agent do to stretch the same spring 4.00 cm from its unstretched position?
9. Four objects are situated along the y axis as follows: a 2.00 kg object is at $+3\text{ m}$, a 3 kg object is at $+2.50\text{ m}$, a 2.50-kg object is at the origin, and a 4 kg object is at -0.5 m . Where is the center of mass of these objects?
10. A billiard ball moving at 5 m/s strikes a stationary ball of the same mass. After the collision, the first ball moves, at 4.33 m/s , at an angle of 30° with respect to the original line of motion. Assuming an elastic collision (and ignoring friction and rotational motion), find the struck ball's velocity after the collision.
11. The center-to-center distance between Earth and Moon is $384\,400\text{ km}$. The Moon completes an orbit in 27.3 days. (a) Determine the Moon's orbital speed. (b) If gravity were switched off, the Moon would move along a straight line tangent to its orbit, as described by Newton's first law. In its actual orbit in 1.00 s , how far does the Moon fall below the tangent line and toward the Earth?

12. Two vectors $\vec{A}$ and $\vec{B}$ (of lengths A and B , respectively) make an angle θ with each other when they are placed tail to tail, see Fig. 2. (a). (a) By taking components along two perpendicular axes, prove that the length of their vector sum $\vec{R} = \vec{A} + \vec{B}$ is

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

- (b) For the difference $\vec{C} = \vec{A} - \vec{B}$, where C is the length of the third side of a triangle formed from connecting the head of $\vec{B}$ to the head of $\vec{A}$ as in Fig. 2. (b), use the same approach to prove that:

$$C = \sqrt{A^2 + B^2 - 2AB \cos \theta}$$

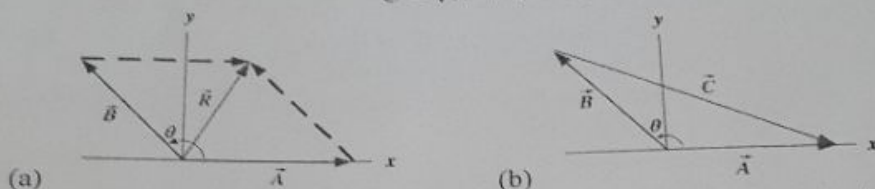


Figure 2 Two vectors $\vec{A}$ and $\vec{B}$ that make an angle θ with each other.

Kinematics

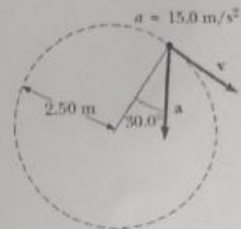
13. Suppose the motion of the particle is described by the equation $x = a + b t^2$, where $a = 10 \text{ m}$ and $b = 2 \text{ m/s}^2$. (a) Find the displacement of the particle in the time interval between $t_i = 2 \text{ s}$ and $t_f = 4 \text{ s}$. (b) Find the average velocity and the average speed during this interval of time.
14. The position of a plane during take-off along a straight runway is given by $x = k t^2$, where $k = 1.2 \text{ m/s}^2$, is measured in meters, and t is in seconds. (a) Find the displacement and the average velocity of the plane in the time intervals $0 \leq t \leq 4 \text{ s}$ and $4 \text{ s} \leq t \leq 10 \text{ s}$. (b) Find the velocity of the plane at $t = 4 \text{ s}$ and at $t = 10 \text{ s}$.
15. An athlete swims the length L of a pool in a time t_1 and makes the return trip to the starting position in a time t_2 . If she is swimming initially in the positive x direction, determine her average velocities symbolically in (a) the first half of the swim, (b) the second half of the swim, and (c) the round trip. (d) What is her average speed for the round trip?
16. An astronaut on a strange planet finds that she can jump a maximum horizontal distance of 15.0 m if her initial speed is 3.00 m/s . What is the free-fall acceleration on the planet?
17. A boy throws a stone horizontally from the top of a cliff of height h toward the ocean below. The stone strikes the ocean at distance d from the base of the cliff. In terms of h , d , and g , find expressions for (a) the time t at which the stone lands in the ocean, (b) the initial speed of the stone, (c) the speed of the stone immediately before it reaches the ocean, and (d) the direction of the stone's velocity immediately before it reaches the ocean.
18. The coordinates of an object moving in the xy plane varying with time according to the equation $x = -5 \sin \omega t$ and $y = 4 - 5 \sin \omega t$ where x and y in meter and t in second and ω is constant.
 - a. Write the expressions for position vector, velocity vector and the acceleration vector at any time t
 - b. Determine the components of the velocity and acceleration and speed at $t = 0 \text{ sec}$.
19. A projectile is fired in such away that its horizontal range is equal to two times its maximum height. What is the angle of the projectile?
20. The velocity of a certain object varies with time according to the equation $\vec{v} = 2\hat{i} + 6t^4\hat{j}$. Obtain the expression for the displacement of the object at any time.
21. The position vector of a certain particle varies with time according to the formula $\vec{r} = \hat{i}e^{-2t} \cos 4t + \hat{j}e^{-3t} \sin 3t + \hat{k}e^{-t}$. At any time calculate velocity and acceleration.

Addis Ababa Science and Technology University

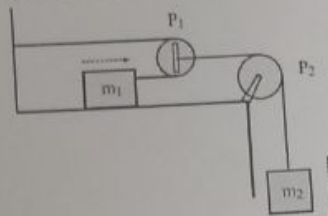
By Tamiru T. (PhD)

Tutorial #1

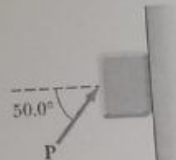
- Figure below represents the total acceleration of a particle moving clockwise in a circle of radius 2.50 m at a certain instant of time. At this instant, find (a) the radial acceleration, (b) the speed of the particle, and (c) its tangential acceleration.



- Mass m_1 on a frictionless horizontal table is connected to mass m_2 through a massless pulley P_1 and a massless fixed pulley P_2 as shown in Figure. (a) If a_1 and a_2 are the magnitudes of the accelerations of m_1 and m_2 respectively, what is the relationship between these accelerations? Find expressions for (b) the tensions in the strings and (c) the accelerations a_1 and a_2 in terms of m_1 , m_2 and g .



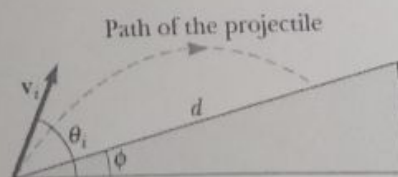
- A block of mass 3 kg is pushed up against a wall by a force P that makes a 50° angle with the horizontal as shown in Figure below. The coefficient of static friction between the block and the wall is 0.250. Determine the possible values for the magnitude of P that allow the block to remain stationary.



- A toy car moving at constant speed completes one lap around a circular track (a distance of 200m) in 25s. (a) What is the average speed? (b) If the mass of the car is 1.50kg, what is the magnitude of the central force that keeps it in a circle?
- A projectile is fired up an incline (incline angle ϕ) with an initial speed v_i at an angle θ_i with respect to the horizontal ($\theta_i > \phi$), as shown in Figure below. (a) Show that the projectile travels a distance d up the incline, where

$$d = \frac{2v_i^2 \cos\theta_i \sin(\theta_i - \phi)}{g \cos^2\phi}$$

- (b) For what value of θ_i is d a maximum, and what is that maximum value?



Wolkite University
College of Natural and Computational Sciences
Department of Physics
General Physics (Phys 1011)
Worksheet I

Uncertainty and significant digits

1. A person walks a distance of $s_1 = 10 \pm 1\text{m}$, then a distance of $s_2 = 5 \pm 1\text{m}$, and finally a distance of $s_3 = 6 \pm 2\text{m}$. Calculate the total distance the person walks?
2. A rectangular plate has a length of $(21.3 \pm 0.2)\text{ cm}$ and a width of $(9.8 \pm 0.1)\text{ cm}$. Calculate the area.
3. Suppose we measure a distance of $(400 \pm 20)\text{ m}$ in $(10 \pm 1)\text{ s}$. Calculate the average speed?
4. The length of a solid cube is 4.00m with uncertainty on length 0.1m . What is the volume of the cube? If the mass of the cube is $12.5 \pm 0.3\text{kg}$, Calculate its density?
5. How many significant figures are in the following numbers? (a) 78.9 ± 0.2 (b) 3.788×10^9 (c) 2.46×10^{26} (d) 0.0053 (e) $0.001\ 223\ 00$
6. Find the result of two significant digits

a. $100.0 + 1.234$	c. 1.234×2.0
b. $12.675 - 6.85$	d. 6.234×3.3454

Vector

7. A car travels 10 km due north and then 5 km due west. Find graphically and analytically the magnitude and direction of the car's resultant displacement.
8. A car travels 6 km due east and then 4 km in a direction 120° north of east. Use both the graphical and analytical methods to find the magnitude and direction of the car's displacement vector.
9. Vector $\vec{A}$ has a magnitude of 10 units and makes 60° with the positive x-axis. Vector $\vec{B}$ has a magnitude of 5 units and is directed along the negative x-axis. Use geometry to find: (a) the vector sum $\vec{A} + \vec{B}$, and (b) the vector difference $\vec{A} - \vec{B}$
10. Three vectors are oriented as shown in Fig.1, where $A=10$, $B=20$, and $C=15$ units. Find: (a) the x and y components of the resultant vector $\vec{D} = \vec{A} + \vec{B} + \vec{C}$, (b) the magnitude and direction of the resultant vector.

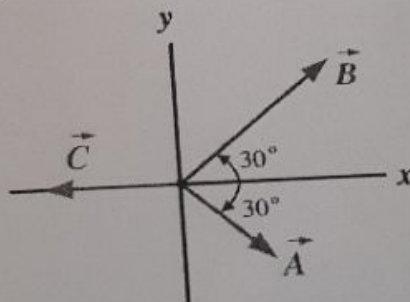
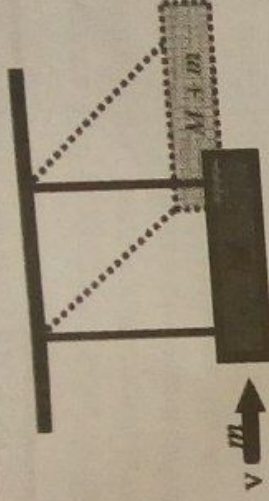


Figure 1. Three vectors on the x-y plane

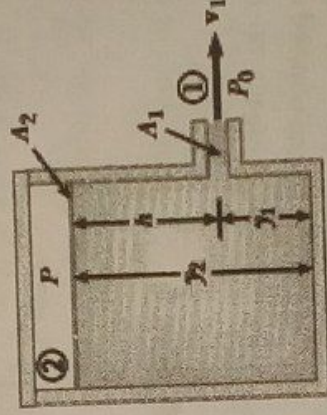
11. Find the vector components of the sum $\vec{R}$ of the displacement vectors $\vec{A}$ and $\vec{B}$? whose components along three perpendicular directions are $A_x=2$, $A_y=1$, $A_z=3$, $B_x=1$, $B_y=4$, and $B_z=2$. Find the magnitude of $\vec{R}$.

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Physics Department Head

General Physics Work Sheet

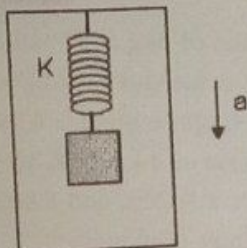


40. Suppose there is a mass of 3kg at $x = 1\text{m}$, a mass of 2 kg at the origin, and a mass of 4 kg at $x = 2\text{ m}$. Where is the center of mass?
41. In two dimensions: suppose there is a mass of 3 kg at $(x, y) = (1, 3)\text{ m}$, a mass of 2 kg at the origin, and a mass of 4 kg at $(x, y) = (5, -1)\text{ m}$. Where is the center of mass?
42. One mass is placed at $x = 2\text{cm}$ and a second mass, equal to the first, is placed at $x = 6\text{cm}$. Where is the center of mass?
43. An enclosed tank containing a liquid of density ρ has a hole in its side at a distance y_1 from the tank's bottom, as shown in figure below. The hole is open to the atmosphere, and its diameter is much smaller than the diameter of the tank. The air above the liquid is maintained at a pressure P . Determine the speed of the liquid as it leaves the hole when the liquid's level is a distance h above the hole.



General Physics Work Sheet

- 1.00 cm when a 12.0 N weight is placed in the pan. When the 12.0 N weight is replaced with a 1.50 kg fish, what distance does the spring stretch?
36. A 10kg block is hanging from a spring with spring constant $K = 1000 \text{ N/m}$. The spring is attached to the ceiling of an elevator. The elevator is currently moving upwards at 10 m/s and slowing down at 1.0 m/s^2 . How much is the spring stretched?



37. A pendulum consists of a sphere of mass m attached to a light cord of length L , as shown in figure below. The sphere is released from rest at point A when the cord makes an angle θ_A with the vertical, and the pivot at P is frictionless.
- A. Find the speed of the sphere when it is at the lowest point B.
- B. What is the tension T_B in the cord at B.

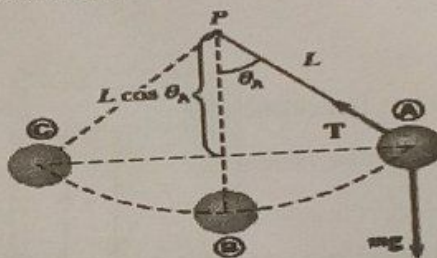
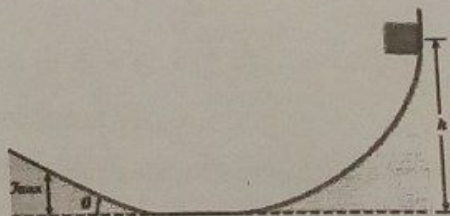


Figure: If the sphere is released from rest at the angle θ_A , it will never swing above this position during its motion.

38. A block slides down a curved frictionless track and then up an inclined plane as shown in figure below. The coefficient of kinetic friction between block and incline is μ_k . Use energy methods to show that the maximum height reached by the block is

$$Y_{\max} = \frac{h}{1 + \mu_k \cot \theta}$$



39. A bullet with mass m is fired into a block of wood with mass M , suspended like a pendulum and makes a completely inelastic collision with it. After the impact, the block swings up to a maximum height y . What is the initial speed of the bullet?

WORK SHEET FOR CHAPTER ONE

1. The magnitudes of two vectors $\vec{A}$ and $\vec{B}$ are $A = 8$ units and $B = 6$ units. Find the largest possible value and smallest possible value.
2. The magnitudes of displacements $\vec{A}$ and $\vec{B}$ are 3 m and 4 m, respectively, and $\vec{C} = \vec{A} + \vec{B}$. Considering various orientations of $\vec{A}$ and $\vec{B}$.
 - ✓ What is the maximum possible magnitude for $\vec{C}$ and
 - ✓ What is the minimum possible magnitude?
3. Vector $\vec{A}$ has x and y components of -8 cm and 12 cm respectively and vector $\vec{B}$ has x and y components of 13 cm and -5 cm respectively. If $\vec{A} - \vec{B} + 2\vec{C} = \vec{0}$ what are the components of $\vec{C}$?
4. Given $\vec{A} = 3\hat{i} + 4\hat{j} - 6\hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + 4\hat{k}$ Find the component of $\vec{A}$ along $\vec{B}$ or the projection of $\vec{A}$ on $\vec{B}$.
5. Determine the coefficients of three vectors. If $\vec{A} = 6\hat{i} - 8\hat{j}$ and $\vec{B} = -8\hat{i} + 3\hat{j}$ and $\vec{C} = 26\hat{i} + 19\hat{j}$. Determine a and b so that $a\vec{A} + b\vec{B} + \vec{C} = \vec{0}$
6. Find the sum, magnitude and unit vector of two vectors. If the given two vectors are $\vec{A} = (3\hat{i} - 2\hat{j} + \hat{k})\text{m}$ and $\vec{B} = (-2\hat{i} + \hat{j} - 2\hat{k})\text{m}$.
 - a. Find $\vec{R} = \vec{A} + \vec{B}$.
 - b. Find the magnitude of $\vec{R}$ and $\hat{R}$.

General Physics Work Sheet

27. For a given pair of surfaces, the maximum value of f_s depends on the normal force n . In some cases the maximum value of f_s is approximately proportional to n . Explain the frictional force for each in figure below.

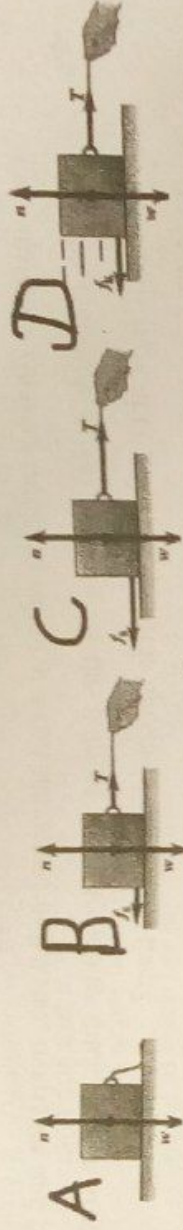
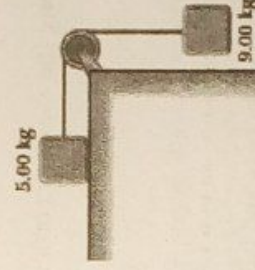


Figure: A) No applied force, box at rest, B) Weak applied force, box remains at rest, C) Stronger applied force, box just start sliding, and D) Box sliding at constant speed.

28. A 5.00kg object placed on a frictionless, horizontal table is connected to a string that passes over a pulley and then is fastened to a hanging 9.00kg object, as in figure below. Draw free-body diagrams of both objects. Find the acceleration of the two objects and the tension in the string.



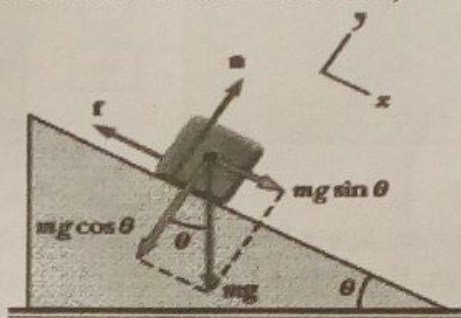
29. When two objects of unequal mass are hung vertically over a frictionless pulley of negligible mass, as in figure below, the arrangement is called an Atwood machine. The device is sometimes used in the laboratory to measure the free-fall acceleration.

- A. Determine the magnitude of the acceleration of the two objects and the tension in the string.
- C. If $m_1 < m_2$, determine the magnitude of the acceleration of the two objects and the tension in the lightweight cord.
- D. If $m_1 > m_2$, determine the magnitude of the acceleration of the two objects and the tension in the lightweight cord.

General Physics Work Sheet



30. Suppose a block is placed on a rough surface inclined relative to the horizontal, as shown in below figure. The incline angle is increased until the block starts to move. Show that by measuring the critical angle θ_c at which this slipping just occurs, we can obtain μ_s (how do we determine the coefficient of friction?).



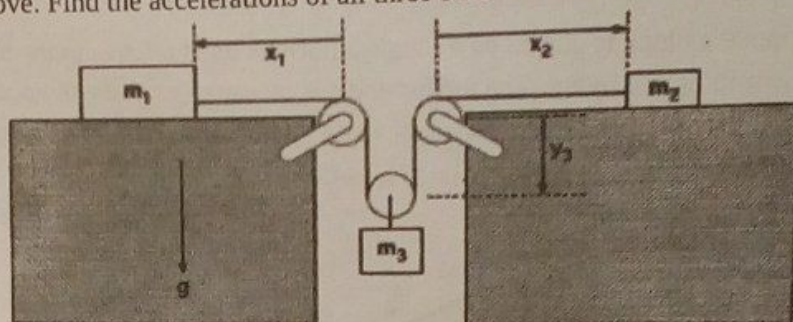
31. A block 1 of mass m_1 , constrained to move along a plane inclined at angle α to the horizontal, is connected via a massless inextensible string that passes over a massless pulley, to a second block 2 of mass m_2 . Assume the coefficient of static friction between the block and the inclined plane is μ_s and the coefficient of kinetic friction is μ_k . Assume the gravitational constant is g .



- A. What is the relation between the masses of block 1 and block 2 such that the system just starts to slip?
 - B. For the following questions suppose block 2 has a mass greater than the value you found in part A.
 - I. Calculate the acceleration of the blocks.
 - II. Calculate the tension in the string.
 - III. Block 2 starts out at a height h above the bottom of the inclined plane and is released at rest. How long does it take to fall a distance S ? Assume that block 1 starts off a distance greater than S from the pulley.
32. In the system shown below $m_1 > m_2$. The pulleys are massless and frictionless, and the rope joining the blocks has no mass. The coefficient of static friction between the blocks and the tables is greater than the coefficient of kinetic friction: $\mu_s > \mu_k$. The downward acceleration of gravity is g .

General Physics Work Sheet

- A. Imagine that when the system is released from rest body 3 accelerates downward at a constant rate of magnitude a , but only one of the other blocks moves. Which block does not move, and what is the magnitude and direction of the friction force holding it back.
- B. Now consider the case where, when released from rest, all three blocks begin to move. Find the accelerations of all three blocks and the tension in the rope.



33. The International Space Station orbit of radius is 6728140m the Earth at an altitude of about 350 km. How fast is it moving?
34. An elevator car has a mass of 1600 kg and is carrying passengers having a combined mass of 200 kg. A constant friction force of 4000 N retards its motion upward, as shown in figure below.
- A. What power delivered by the motor is required to lift the elevator car at a constant speed of 3.00 m/s?
- B. What power must the motor deliver at the instant the speed of the elevator is v if the motor is designed to provide the elevator car with an upward acceleration of $1.00 \frac{m}{s^2}$?

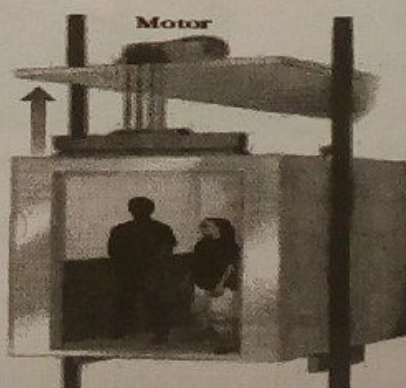


Figure: The motor exerts an upward force T on the elevator car. The magnitude of this force is the tension T in the cable connecting the car and motor. The downward forces acting on the car are a friction force f and the gravitational force $F_g = Mg$.

35. In this example we will consider the forces that come into play in a spring balance. Suppose that a spring balance used to weigh fish is built with a spring that stretches

General Physics Work Sheet

9. A car is driven east for a distance of 50 km, then north for 30 km, and then in a direction 30° east of north for 25 km. Sketch the vector diagram and determine (a) the magnitude and (b) the angle of the car's total displacement from its starting point.
10. A person desires to reach a point that is 3.40 km from her present location and in a direction that is 35.0° north of east. However, she must travel along streets that are oriented either north-south or east-west. What is the minimum distance she could travel to reach her destination?
11. An airplane flying horizontally at a constant speed of 350 km/h over level ground releases a bundle of food supplies. Ignore the effect of the air on the bundle. What are the bundle's initial (a) vertical and (b) horizontal components of velocity? (c) What is its horizontal component of velocity just before hitting the ground? (d) If the airplane's speed were, instead, 450 km/h, would the time of fall be longer, shorter, or the same?
12. Is it possible to be accelerating while traveling at constant speed? Is it possible to round a curve with (b) zero acceleration and (c) a constant magnitude of acceleration?
13. While riding in a moving car, you toss an egg directly upward. Does the egg tend to land behind you, in front of you, or back in your hands if the car is (a) traveling at a constant speed, (b) increasing in speed, and (c) decreasing in speed?
14. A snowball is thrown from ground level (by someone in a hole) with initial speed v_0 at an angle of 45° relative to the (level) ground, on which the snowball later lands. If the launch angle is increased, do (a) the range and (b) the flight time increase, decrease, or stay the same?
15. You are driving directly behind a pickup truck, going at the same speed as the truck. A crate falls from the bed of the truck to the road. (a) Will your car hit the crate before the crate hits the road if you neither brake nor swerve? (b) During the fall, is the horizontal speed of the crate more than, less than, or the same as that of the truck?
16. At what point in the path of a projectile is the speed a minimum?
17. In shot put, the shot is put (thrown) from above the athlete's shoulder level. Is the launch angle that produces the greatest range 45° , less than 45° , or greater than 45° ?
18. A stone is dropped at $t = 0$. A second stone, with twice the mass of the first, is dropped from the same point at $t = 100$ ms. (a) How far below the release point is the center of mass of the two stones at $t = 300$ ms? (Neither stone has yet reached the

General Physics Work Sheet

- ground.) (b) How fast is the center of mass of the two stone systems moving at that time?
19. A 1000 kg automobile is at rest at a traffic signal. At the instant the light turns green, the automobile starts to move with a constant acceleration of 4.0 m/s^2 . At the same instant a 2000 kg truck, traveling at a constant speed of 8.0 m/s , overtakes and passes the automobile. (a) How far is the com of the automobile-truck system from the traffic light at $t = 3.0 \text{ s}$? (b) What is the speed of the com then?
20. A 91 kg man lying on a surface of negligible friction shoves a 68 g stone away from himself, giving it a speed of 4.0 m/s . What speed does the man acquire as a result?
21. A space vehicle is traveling at 4300 km/h relative to Earth when the exhausted rocket motor (mass $4m$) is disengaged and sent backward with a speed of 82 km/h relative to the command module (mass m). What is the speed of the command module relative to Earth just after the separation?
22. A particle moves along the x-axis according to the equation $x(t) = 2 + 3t - t^2$, where t is
- II. The displacement between $t = 1\text{s}$ and $t = 3\text{s}$.
 - III. Average velocity between $t = 1\text{s}$ and $t = 3\text{s}$.
 - IV. Instantaneous.
23. A projectile is fired in such a way that its horizontal range is equal to three times its maximum height. What is the angle of projection?
24. Two projectiles are thrown with the same initial velocity, one at an angle θ and the other at an angle of $90^\circ - \theta$.
- I. Can both projectiles strike the ground at the same distance from the projection point?
 - II. Can both projectiles be in air for the same time interval?
25. Suppose the muzzle velocity is $V_0 = 40 \text{ m/s}$ and the target is at a distance of $R = 75 \text{ m}$. What launch angle is needed to hit the target?
26. Suppose the muzzle velocity is $V_0 = 40 \text{ m/s}$ and the target is at a distance of $R = 200 \text{ m}$. What launch angle is needed to hit the target?

General Physics Work Sheet

Debre Berahan University

College of Natural and Computational Sciences

Department of Physics(General Physics Work Sheet for YEAR I Students)

1. Describe two vectors $\vec{a}$ and $\vec{b}$ such that
 - (a) $\vec{a} + \vec{b} = \vec{c}$ and $a + b = c$;
 - (b) $\vec{a} + \vec{b} = \vec{a} - \vec{b}$;
 - (c) $\vec{a} + \vec{b} = \vec{c}$ and $a^2 + b^2 = c^2$
2. Two vectors are given by
$$\vec{a} = (4.0 \text{ m})\hat{i} - (3.0 \text{ m})\hat{j} + (1.0 \text{ m})\hat{k} \text{ and } \vec{b} = (-1.0 \text{ m})\hat{i} + (1.0 \text{ m})\hat{j} + (4.0 \text{ m})\hat{k}.$$
In unit-vector notation, find
 - (a) $\vec{a} + \vec{b}$,
 - (b) $\vec{a} - \vec{b}$, and
 - (c) a third vector $\vec{c}$ such that $\vec{a} - \vec{b} + \vec{c} = 0$.
3. What are (a) the x component and (b) the y component of a vector in the xy plane if its direction is 250° counter clockwise from the positive direction of the x axis and its magnitude is 7.3 m?
4. A person walks 57.0 m at 47.0° north of east, turns and walks 72.0 m at 15.0° south of east, and then turns and walks 24.0 m 30.0° west of north.
 - A. How far and at what angle is the person's final position from his/her initial position?
 - B. In what direction would the person have to head to return to his/her initial position?
5. Find the vector projection of $\vec{u} = 6\hat{i} + 3\hat{j} + 2\hat{k}$ on to $\vec{v} = \hat{i} - 2\hat{j} - 2\hat{k}$ and the scalar component of $\vec{u}$ in the direction of $\vec{v}$.
6. If the vectors $(k, 2, 4)$ and $(k, 5k, 4)$ are orthogonal. Find the value of k.
7. The x component of vector $\vec{A}$ is -25.0 m and the y component is +40.0 m.
 - (a) What is the magnitude of $\vec{A}$?
 - (b) What is the angle between the direction of $\vec{A}$ and the positive direction of x?
8. A ship sets out to sail to a point 120 km due north. An unexpected storm blows the ship to a point 100 km due east of its starting point. (a) How far and (b) in what direction must it now sail to reach its original destination?

Bahir Dar University, Department of Physics

General Physics (Phys 1011)

Work sheet 1

Contributed by: Andargie G. , Abiy T. , Meseret L., Mekdes M. and Fresenay.M

Useful constants: $g = 9.8 \text{ m/s}^2$.

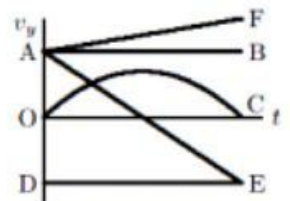
Part I: Choose the best answer(s) from the given alternatives

1. A stone that is dropped freely from rest traveled half the total height in the last second. With what Velocity will it strike the ground?

A) 34 m/s B) 44 m/s C) 47 m/s D) 5.7 m/s E) None

2. Which of the curves on the graph below best represents the vertical component of the velocity versus the time for a projectile fired at an angle of 45° above the horizontal?

A) OC B) DE C) AB D) AE E) AF



3. A bullet is dropped from the same height when another bullet is fired horizontally. They will hit the ground:

A) One after the other B) Simultaneously C) Depends on the observer D) None of the above

4. Coin I is thrown upward from the top of a 100-m tower with a speed of 15 m/s. Coin II is dropped from the top of the tower 2.0 s later. Assume g is 10 m/s^2 . How far below the top of the tower does coin I pass coin II?

A) 1.6 m B) 16 m C) 20 m D) 80 m E) 96 m

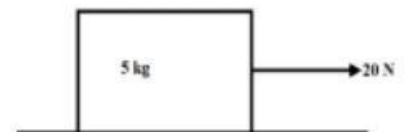
5. The total mass of an elevator with a 80.0kg man in it is $1.0 \times 10^3 \text{ kg}$. This elevator moving upward with a speed of 8.0 m/s, is brought to rest over a distance of 20.0 m.

(I) the tension in the cables supporting the elevator

A) 4100 N B) 6272 N C) 8200 N D) 784 N E) None

(II) The force exerted on the man by the elevator floor.

A) 112 N B) 656 N C) 328 N D) 224 N E) None



6. A block is pulled across a horizontal surface as shown. The mass of the block is 5.0 kg. The block is travelling at a constant velocity. What is the calculated force of friction acting on the block?

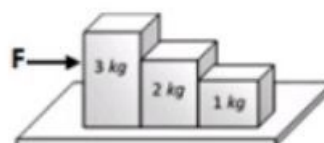
A) 0 N B) 4 N C) 15 N D) 20 N E) 25 N

7. A wooden block slides down a 45° inclined plane in twice the time it takes to slide down a frictionless 45° inclined plane. What is the coefficient of kinetic friction between block and inclined plane?

A) 0.25 B) 0.5 C) 0.6 D) 0.75 E) None

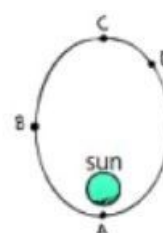
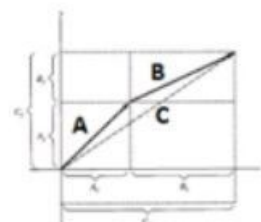
8. Consider the following statements about the blocks shown in the diagram that are being pushed by a constant force on a frictionless table

- i. All blocks move with the same acceleration
- ii. The net force on each block is the same.



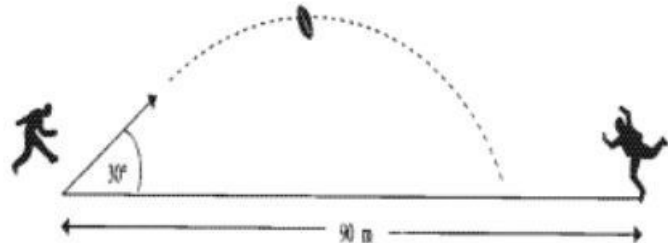
Which of these statements are/is correct

- A) i only B) ii only C) Both i and ii D) Neither i nor ii
9. Which of the following is true about the force of gravity between two objects?
- A) It increases linearly with the masses, but decreases linearly with their separation.
 - B) It increases linearly with the masses, but decreases with the square of their separation.
 - C) It increases with the cube of their masses, but decreases linearly with their separation.
 - D) It decreases linearly with the masses, and decreases linearly with their separation.
 - E) It decreases linearly with the masses, but increases with the square of their separation.
10. If our planet Earth were orbiting the Sun in an orbit four times as big as its current orbit, how many times longer would it take it to go around the Sun than it does now?
- A) 4 B) 8 C) 16 D) 24
11. A bullet is fired from a rifle. If the rifle recoils freely, then the kinetic energy of the rifle is
- A) Less than that of the bullet
 - B) More than that of the bullet
 - C) Same as that of the bullet
 - D) Equal or less than that of the bullet
12. If a projectile thrown vertically upward rises to a height of 500.0m, what is the range and the velocity it was thrown?
- A) 0 m & $9.9 \frac{m}{s}$ B) 0 m & $99 \frac{m}{s}$ C) 0 m & $9.8 \times 10^3 \frac{m}{s}$ D) 9.9 m & $9.9 \frac{m}{s}$
13. Which one of the following is not true about deceleration?
- A) The magnitude of the velocity is decreasing.
 - B) The velocity and acceleration point in opposite directions.
 - C) The acceleration is necessarily negative.
 - D) all
14. A 50.0 kg wooden block is placed on a concrete floor. If the coefficient of friction between the surfaces is 0.600, what is the magnitude of the minimum horizontal force needed to set the block in motion?
- A) 490 N B) 24 N C) 750 N D) 294 N
15. Assume vector **A** has a magnitude **A** = 7.0 unit and makes an angle of 40 degree with the x axis. Vector **B** = 8.0 unit and makes an angle of 30 degree with the x axis as shown in the figure. What is the magnitude and direction of vector **C** = **A** + **B**.
- A) 15.0 and 34.6° B) 10.0 and 30° C) 12.0 and 43.6° D) 11.0 and 24.6°
16. The orbital path of a satellite is shown in the sketch. In which marked positions A through D does the satellite have the greatest KE ?
- A) at A B) at B C) at C D) at D



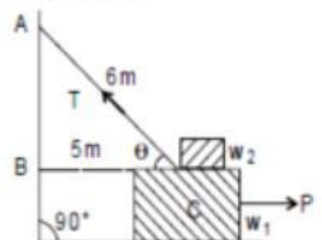
Part II: Short Answer (Write your answer without showing steps with proper significant and decimal digits)

- The position of an object is related to time by $x = t^3 - 6t^2 - 15t + 40$. Where x in meter. Find
 - The Displacement between the time interval from $t = 4$ to $t = 6$ seconds
 - the time at which the velocity will be zero.
 - The instantaneous velocity and acceleration as functions of time
- Vector **A** has x and y components of -8.70 cm and 15.0 cm, respectively; vector **B** has x and y components of 13.2 cm and -6.60 cm, respectively. If $\mathbf{A} - \mathbf{B} + 3\mathbf{C} = 0$, what are the components of **C**?
- A sailor in a small sailboat encounters shifting winds. She sails 2.00 Km east, then 3.5 Km 45° S of E, and then an additional distance in an unknown direction. Her final position is 5.80 Km directly east of the starting point. Find the magnitude and direction of the third leg of the journey using component method.
- Two stones are dropped from the edge of a 60 -m cliff, the second stone 1.6 s after the first. How far below the top of the cliff is the second stone when the separation between the two stones is 36 m?
- A footballer kicks a football so that it leaves the ground at an angle of 30° with a velocity of 30 m/s. The ball is aimed directly at an opposing player 90 m away who, the moment the ball is kicked, starts to run forward to catch the ball (see Figure to the right). At what speed must he run to just catch the ball?
- With what horizontal velocity must a satellite be projected at 800 km above the surface of the earth so that it will have a circular orbit about the Earth? Assume Earth's radius 6400 km. What will be the period of rotation?
- The period of a satellite in circular orbit of radius 12000 km around a planet is 3 hours. Obtain the period of a satellite in circular orbit of radius 48000 km around the same planet.
- A man pulls a block weighing 10 kg up to a distance of 10 meter on a horizontal rough surface (coefficient of kinetic friction $= 0.20$) at a constant speed. Calculate the work done by the man if the pull exerted by him makes an angle of 30° with the horizontal.



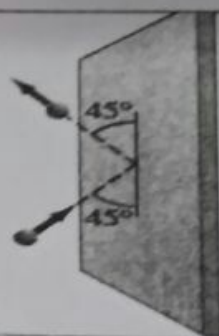
Problems for assignment (8 students in a group) to be submitted after Mid Exam.

- Determine the percentage of kinetic energy lost by a small ball when it makes an elastic head-on collision with stationary bigger ball. The mass of the bigger ball is 12 times bigger than the mass of the small ball.
- A block weighing 400 kg rests on a horizontal surface and supports on top of it, another block of weight 100 kg as shown. The block W_2 is attached to a vertical wall by a string 6 m long. Find the magnitude of the horizontal force P applied to the lower block as shown in figure that shall be necessary so that slipping of W_2 occurs.
- A sphere of mass m , moving with velocity V , enters a hanging bag of sand and stops. If the mass of the bag is M and it is raised by height h , express the Velocity of the sphere (V) in terms of m , M and h .



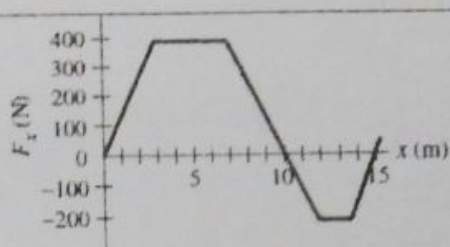
- 0.380-kg ball when released? (b) How high above its original position (spring compressed) will the ball fly?
20. A pendulum 2.00 m long is released (from rest) when the string makes an angle of $\theta_0 = 30.0^\circ$ with vertical. Determine the speed of the 70.0-g bob: (a) at the lowest point (b) at $\theta = 15.0^\circ$ (c) at $\theta = -15.0^\circ$ (i.e., on the opposite side). (d) Determine the tension in the cord at each of these three points. (e) If the bob is given an initial speed when released at $\theta_0 = 30.0^\circ$, recalculate the speeds for parts (a), (b), and (c).
21. An object at rest is suddenly broken apart into two fragments by an explosion. One fragment acquires twice the kinetic energy of the other. What is the ratio of their masses?
22. A 22-g bullet traveling 210 m/s penetrates a 2.0-kg block of wood and emerges going 150 m/s. If the block is stationary on a frictionless surface when hit, how fast does it move after the bullet emerges?
23. A 12-kg hammer strikes a nail at a velocity of 8.5 m/s and comes to rest in a time interval of 8.0 ms. (a) What is the impulse given to the nail? (b) What is the average force acting on the nail?

24. A tennis ball of mass $m = 0.060$ kg and speed $v = 25$ m/s strikes a wall at a 45° angle and rebounds with the same speed at 45° as shown. What is the impulse (magnitude and direction) given to the ball?

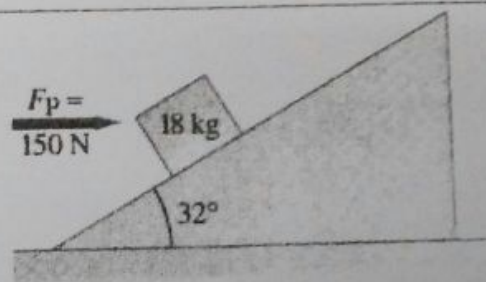


25. A 0.450-kg hockey puck, moving east with a speed of 4.80 m/s has a head-on collision with a 0.900-kg puck initially at rest. Assuming a perfectly elastic collision, what will be the speed and direction of each object after the collision?
26. A 10 g bullet is fired into a 10 kg wood block that is at rest on a wood table. The block, with the bullet embedded, slides 5.0 cm across the table. What was the speed of the bullet?
27. A 10 g snow ball traveling east at 3.0 m/s collides with and sticks to a 20 g snow ball traveling 45° south of west at 2 m/s. What is the speed of the resulting snowball?
28. Three objects lie in the x-y plane: 1.40 kg at (3.9 m, 9.56 m) ; 1.90 kg at (-6.58 m, -15.6 m) and 2.40 kg at (0 m, -14.4 m). What are the center of mass coordinates?

13. The force on a particle, acting along the x axis, varies as shown in the diagram. Determine the work done by this force to move the particle along the x axis: (a) from $x = 0.00$ to $x = 10.0$ m; (b) from $x = 0.0$ to $x = 15.0$ m.



14. (a) How much work is done by the horizontal force on the 18-kg block (see the figure) when the force pushes the block 5.0 m up along the 32° ? (b) How much work is done by the gravitational force on the block during this displacement? (c) How much work is done by the normal force? (d) What is the work done by friction? (e) What is the speed of the block (assume that it is zero initially) after this displacement? The coefficient of kinetic friction between the block and the incline is $\mu_k = 0.01$.

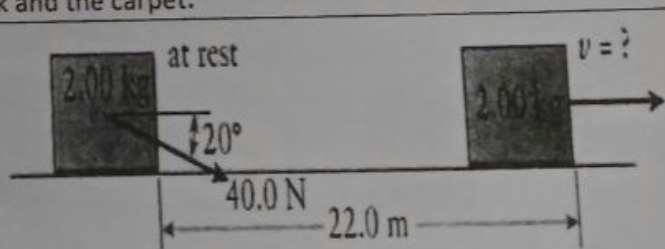


15. At an accident scene on a level road, investigators measure a car's skid mark to be 98 m long. It was a rainy day and the coefficient of friction was estimated to be 0.38. Use these data to determine the speed of the car when the driver slammed on (and locked) the brakes. (Why does the car's mass not matter?)
16. 46.0-kg crate, starting from rest, is pulled across a floor with a constant horizontal force of 225 N. For the first 11.0 m the floor is frictionless, and for the next 10.0 m the coefficient of friction is 0.20. What is the final speed of the crate after being pulled these 21.0 m?
17. A 1.60-m tall person lifts a 1.95-kg book off the ground so it is 2.20 m above the ground. What is the potential energy of the book relative to (a) the ground, and (b) the top of the person's head? (c) How is the work done by the person related to the answers in parts (a) and (b)?
18. In the high jump, the kinetic energy of an athlete is transformed into gravitational potential energy without the aid of a pole. With what minimum speed must the athlete leave the ground in order to lift his center of mass 2.10 m and cross the bar with a speed of 0.70 m/s?
19. A vertical spring (ignore its mass), whose spring constant is 875 N/m is attached to a table and is compressed down by 0.160 m. (a) What upward speed can it give to a

Problems

1. A hypothetical planet has a radius 2.3 times that of Earth, but has the same mass. What is the acceleration due to gravity near its surface?
2. A hypothetical planet has a mass 1.80 times that of Earth, but the same radius. What is g near its surface?
3. If you doubled the mass and tripled the radius of a planet, by what factor would g at its surface change?
4. Calculate the effective value of g , the acceleration of gravity, at (a) 6400 m, and (b) 6400 km, above the Earth's surface.
5. Four 8.5-kg spheres are located at the corners of a square of side 0.80 m. Calculate the magnitude and direction of the gravitational force exerted on one sphere by the other three.
6. Two objects attract each other gravitationally with a force of $2.5 \times 10^{-10} \text{ N}$ when they are 0.25 m apart. Their total mass is 4.00 kg. Find their individual masses.
7. Calculate the speed of a satellite moving in a stable circular orbit about the Earth at a height of 5800 km.
8. Two satellites orbit Earth at altitudes of 5000 km and 15,000 km. Which satellite is faster, and by what factor?
9. A 350-kg box is pulled 7.00 m up a 30.0° inclined plane by an external force of $5.00 \times 10^3 \text{ N}$ that acts parallel to the frictionless plane. Calculate the work done by (a) the external force, (b) gravity, and (c) the normal force.
10. A 1250-kg car moves at 20.0 m/s. How much work must be done on the car to increase its speed to 30.0 m/s?
11. A book slides across a level, carpeted floor at an initial speed of 4.00 m/s and comes to rest after 3.25 m. Applying the work-energy theorem, calculate the coefficient of kinetic friction between the book and the carpet.

12. Calculate the final speed of the 2.00-kg object that is pushed for 22.0 m by a 40.0-N force directed 20.0° below the horizontal on a level, frictionless floor (see the figure). Assume the object starts from rest.



direction of the force? (b) Determine the time it takes for the particle to come to a stop. (c) What is its mass of the particle?

20. Two blocks of mass 4.0 and 6.0 kg are joined by a string and rest on a frictionless horizontal table. If a force of 100 N is applied horizontally on one of the blocks, find the acceleration of each block.

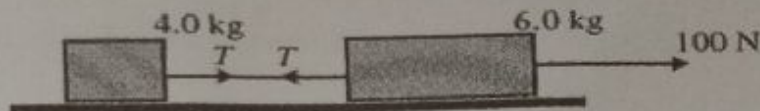


Figure: 2

21. A 455 kg crate is being pulled at constant velocity by a force directed at 30° to the horizontal as shown in Fig.3 below. The frictional force on the crate is 1163 N. What is the magnitude of the pulling force?

22. An ice boat is coasting along a frozen lake. Friction between the ice and the boat is negligible, and so is air resistance. Nothing is propelling the boat. From a bridge, someone jumps straight down and lands in the boat, which continues to coast straight ahead.

(a) Does the horizontal momentum of the boat change?

(b) Does the speed of the boat increase, decrease, or remain the same? Explain your answers.

23. Two objects with masses of 3.00 kg and 5.00 kg are connected by a light string that passes over a frictionless pulley, as shown following Figure 4. Determine (a) the tension in the string, b) the acceleration of each object.

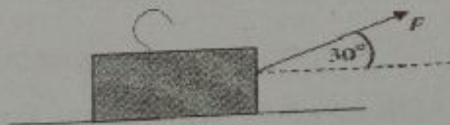


Figure: 3

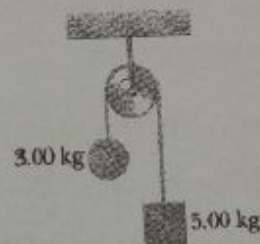
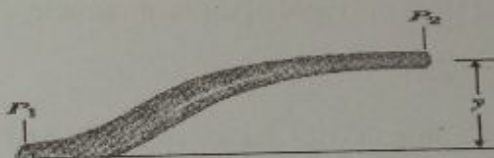


Figure: 4

24. A certain truck has twice the mass of a car. Both are moving at the same speed. If the kinetic energy of the truck is K , what is the kinetic energy of the car? (a) $K/4$ (b) $K/2$ (c) $0.71K$ (d) $2K$
25. If two objects collide and one is initially at rest, is it possible for both to be at rest after the collision? Is it possible for one to be at rest after the collision? Explain.
26. The normal force acting on an object and the force of static friction do zero work on the object. However, the reason that the work is zero is different for the two cases. Explain why each does zero work.
27. A hydraulic jack has an input piston of area 0.050m^2 and an output piston of area 0.70m^2 . How much force on the input piston is required to lift a car weighing 1200N ?
28. A 2.0kg particle has a velocity of $\mathbf{v}_1 = (2\mathbf{i} - 3\mathbf{j})\text{ m/s}$, and a 3.0kg particle has a velocity $\mathbf{v}_2 = (1.0\mathbf{i} + 6.0\mathbf{j})\text{ m/s}$. Find (a) the velocity of the center of mass and (b) the total momentum of the system.
29. 30. A 40-cm tall glass is filled with water to a depth of 30 cm . A) What is the gauge pressure at the bottom of the glass? B) What is the absolute pressure at the bottom of the glass?
30. 31. Water moves through a constricted pipe in steady, ideal flow. At the lower point shown in Figure 5 below, the pressure is 175000Pa and the pipe radius is 3.00 cm . At the higher point located at $y = 2.50\text{m}$, the pressure is 120000 Pa and the pipe radius is 1.50 cm . Find the speed of flow
- 
- a) In the lower section b) In the upper section (c) Find the volume flow rate through the pipe.
31. If the pressure reading of your pitot tube is 15.0 mm Hg at a speed of 200 km/h , what will it be at 700 km/h at the same altitude?
32. A billiard ball moving at 5.00 m/s strikes a stationary ball of the same mass. After the collision, the first ball moves at 4.33 m/s at an angle of 30.0° with respect to the original line of motion. Assuming an elastic collision (and ignoring friction and rotational motion), find the struck ball's velocity.

Addis Ababa University
College of Natural and Computational Sciences
Department of Physics
Phys 1011 Worksheet II

Conceptual Questions

1. If the acceleration of an object is zero, are no forces acting on it? Explain.
2. If an object is moving, is it possible for the net force acting on it to be zero?
3. Only one force acts on an object. Can the object have zero acceleration? Can it have zero velocity? Explain.
4. If you walk along a log floating on a lake, why does the log move in the opposite direction?
5. The force of gravity on a 2-kg rock is twice as great as that on a 1-kg rock. Why then doesn't the heavier rock fall faster?
6. You pull a box with a constant force across a frictionless table using an attached rope held horizontally. If you now pull the rope with the same force at an angle to the horizontal (with the box remaining flat on the table), does the acceleration of the box (a) remain the same, (b) increase, or (c) decrease? Explain.
7. Which pulls harder gravitationally, the Earth on the Moon, or the Moon on the Earth? Which accelerates more?
8. Would it require less speed to launch a satellite (a) toward the east or (b) toward the west? Consider the Earth's rotation direction.
9. Kepler's laws tell us that a planet moves faster when it is closer to the Sun than when it is farther from the Sun. What causes this change in speed of the planet?
10. Can a centripetal force ever do work on an object? Explain.
11. Can the normal force on an object ever do work? Explain.
12. You have two springs that are identical except that spring 1 is stiffer than spring 2 ($k_1 > k_2$). On which spring is more work done: (a) if they are stretched using the same force; (b) if they are stretched the same distance?
13. If the speed of a particle triples, by what factor does its kinetic energy increase?
14. Two bullets are fired at the same time with the same kinetic energy. If one bullet has twice the mass of the other, which has the greater speed and by what factor? Which can do the most work?
15. A light object and a heavy object have the same kinetic energy. Which has the greater momentum? Explain.
16. A glass will break if it falls onto a hardwood floor but not if it falls from the same height onto a padded, carpeted floor. Explain why.
17. One ball has four times the mass and twice the speed of another. (a) How does the magnitude of momentum of the more massive ball compare to the magnitude of momentum of the less massive one? (b) How does the kinetic energy of the more massive ball compare to the kinetic energy of the less massive one?

13. The driver of a car steps on the brakes, and the velocity drops from 20 m/s due east to 10 m/s due east in a time of 2.0 seconds. What is the acceleration?

14. Find the components of the vectors shown in figure 1 below. The magnitude of $\vec{a}$ is 10 units and that of $\vec{b}$ is 20 units.

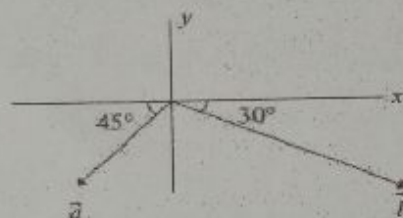


Figure:1

15. In a football game a kicker attempts a field goal. The ball remains in contact with the kicker's foot for 0.050 s, during which time it experiences an acceleration of 340 m/s^2 . The ball is launched at an angle of 51° above the ground. Determine the horizontal and vertical components of the launch velocity.

16. Two ice skaters, Megan and Jason, push off from each other on frictionless ice. Jason's mass is twice that of Megan. (a) Which skater, if either, experiences the greater impulse during the push? Explain. (b) Which skater, if either, has the greater speed after the push off? Explain.

17. A fish swimming in a horizontal plane has a velocity $\vec{v}_0 = (4.0\hat{i} + 1.0\hat{j}) \text{ m/s}$ at a point in the ocean whose position vector is $\vec{r}_0 = (10.0\hat{i} - 4.0\hat{j}) \text{ m}$ relative to a stationary rock at the shore. After the fish swims with constant acceleration for 20.0 s, its velocity is $\vec{v} = (20.0\hat{i} - 5.0\hat{j}) \text{ m/s}$. (a) what are the components of the acceleration? (b) What is the direction of the acceleration with respect to the fixed x axis? (c) Where is the fish at $t = 25 \text{ s}$ and in what direction is it moving?

18. The coefficient of static friction between a 40.0 kg picnic table and the ground is below it is 0.43 m. What is the greatest horizontal force that could be exerted on the table while it remains stationary?

19. A particle is traveling in a straight line at a constant speed of 25.0 m/s. suddenly, a constant force of 15.0 N acts on it, bringing it to a stop in a distance of 62.5 m. (a) what is the

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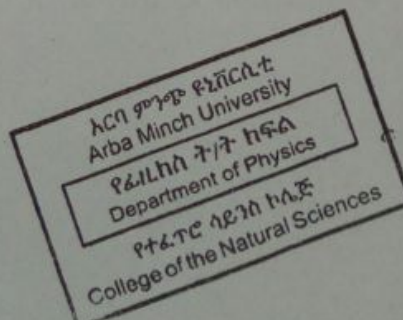
Department of Physics

General Physics (Phys 1011) - Worksheet I

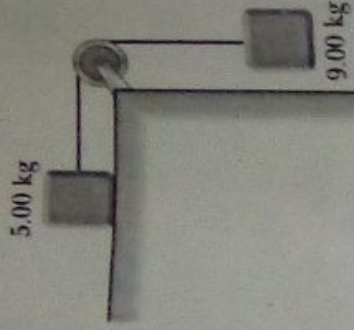
1. What are the primary sources of uncertainty of measurements?
2. Define uncertainty of measurement and explain sources of uncertainty in measurement.
3. In scientific measurement why do always the last digits subjected to uncertainty of measurements?
4. How many significant digits do the following numbers have if it is expressed in scientific notation?
a) 0.000011111000 kg b) 2000000m c) 10000.000022222 d) 0.0001kg X 0.45002222kg
5. Express the following quantities using the prefixes and the unit abbreviations.
a) 1 000 000 watts, b) 0.002 gram, c) 3×10^6 meters, d) 30 000 seconds.
6. Show that the product of mass, acceleration, and speed have the dimensions of power.
7. How many inches are in 30.7484 miles?
8. Can you find two vectors with different lengths that have a vector sum of zero? What length restrictions are required for three vectors to have a vector sum of zero? Explain
9. Find a unit vector in the direction of the resultant of vectors

$$\vec{A} = 2i - 3j + k, \vec{B} = i + j + k, \vec{C} = 2i - 3j + k$$

10. A fish swimming in a horizontal plane has a velocity $v_0 = (4.0i + 1.0j)$ m/s at a point in the ocean whose position vector is $r_0 = (10.0i - 4.0j)$ m relative to a stationary rock at the shore. After the fish swims with constant acceleration for 20.0 s, its velocity is $v = (20.0i - 5.0j)$ m/s. (a) what are the components of the acceleration? (b) What is the direction of the acceleration with respect to the fixed x axis? (c) Where is the fish at $t = 25$ s and in what direction is it moving?
11. A baseball is tossed into the air. After 0.30s it is at the position (2.2 m, 2.5 m). After 1.3 s it is at the position (9.6 m, 4.5 m). Find the magnitude and direction of the displacement vector.
12. A force of 5.0 N is applied to a mass of 10 kg. If the mass starts from rest, how far will the force move in 10 seconds?



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35. Indicate how many significant figures there are in each of the following measured values.

246.32	107.854	100.3	0.678	1.008
0.00340				
14.600	0.0001	700000	350.670	1.000

36. Calculate the answer to the appropriate number of significant figures.

a) 23.7×3.8	=		e) 43.678×64.1	=	
b) 45.76×0.25	=		f) $1.678 / 0.42$	=	
c) 81.04×0.010	=		g) $28.367 / 3.74$	=	
d) 6.47×64.5	=		h) $4278 / 1.006$	=	



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15. Which of the following identities is not valid?
 - (a) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
 - (b) $\mathbf{a}(\mathbf{b} + \mathbf{c}) = \mathbf{ab} + \mathbf{bc}$
 - (c) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
 - (d) $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = -\mathbf{b} \cdot (\mathbf{a} \times \mathbf{c})$
 - (e) $\mathbf{a}_n \cdot \mathbf{a}_b = \cos \theta_{ab}$
16. Let $\mathbf{A} = 2\mathbf{a}_x + 5\mathbf{a}_y - 3\mathbf{a}_z$, $\mathbf{B} = 3\mathbf{a}_x - 4\mathbf{a}_y$, and $\mathbf{C} = \mathbf{a}_x + \mathbf{a}_y + \mathbf{a}_z$.
 - (a) Determine $\mathbf{A} + 2\mathbf{B}$
 - (b) Calculate $|\mathbf{A} - 5\mathbf{C}|$
 - (c) For what value of k is $|\mathbf{KB}| = 2$?
 - (d) Find $(\mathbf{A} \times \mathbf{B}) / (\mathbf{A} \cdot \mathbf{B})$
17. If $\mathbf{A} = 5\mathbf{a}_x + 3\mathbf{a}_y + 2\mathbf{a}_z$, $\mathbf{B} = -\mathbf{a}_x + 4\mathbf{a}_y + 6\mathbf{a}_z$ and $\mathbf{C} = 8\mathbf{a}_x + 2\mathbf{a}_y$ find the value of α and β such that $\alpha\mathbf{A} + \beta\mathbf{B} + \mathbf{C}$ is parallel to the y -axis.
18. Show that $(\mathbf{A} \cdot \mathbf{B})^2 + (\mathbf{A} \times \mathbf{B})^2 = A^2 B^2$
19. In a given vector $\mathbf{A} = \alpha\mathbf{a}_x + \mathbf{a}_y + 4\mathbf{a}_z$, $\mathbf{B} = 3\mathbf{a}_x + \beta\mathbf{a}_y - 6\mathbf{a}_z$ and $\mathbf{C} = 5\mathbf{a}_x - 2\mathbf{a}_y + \gamma\mathbf{a}_z$ determine α , β and γ such that the vectors are mutually orthogonal.
20. Show that the dot and cross in the triple scalar product may be interchanged that is

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$$
21. How many grams are in 5 pounds?
22. You are traveling in Europe and want to know how fast to drive to maximize fuel efficiency. The optimal driving speed for fuel efficiency is 55 miles per hour. How fast would that be in kilometers per hour?
23. In the equation $mgh = \frac{1}{2}mv^2$, h represents distance, g represents acceleration due to gravity, m represents mass and v represents velocity. Show that the dimensions of both sides of the equation match up.
24. A Top Fuel dragster can accelerate from 0 to 100 mph (160 km/hr) in 0.8 seconds. What is the average acceleration in m/s^2 ?
25. While driving through Napa you observe a hot air balloon in the sky with tourists on board. One of the passengers accidentally drops a wine bottle and you note that it takes 2.3 seconds for it to reach the ground.
 - (a) How high is the balloon?
 - (b) What was the wine bottle's velocity just before it hit the ground?
26. A tennis ball is launched 32° above the horizontal at a speed of 7.0 m/s. What are the horizontal and vertical velocity components?

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27. A golf ball was hit into the air with an initial velocity of 4.47 m/s at an angle of 66° above the horizontal. How high did the ball go and how far did it fly horizontally?

28. A 0.15 kg baseball is thrown horizontally at 40 m/s and after it is struck by a bat, it is traveling at -40 M/s.

(a) what impulse did the bat deliver to the ball?

(c) If the contact time of the bat and ball was 0.00080 seconds, what was the average force the bat exerted on the ball?

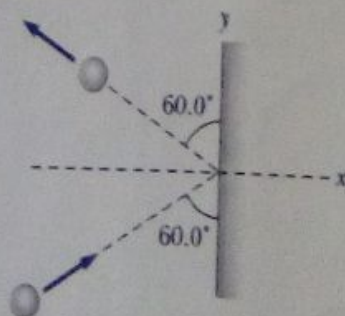
(d) Calculate the average acceleration of the ball during the time it was in contact with the bat.

We can calculate the change in momentum and give the answer as impulse because we know that the impulse is equal to the change in momentum.

29. (a) A particle of mass m moves with momentum p . Show that the kinetic energy of the particle is $K = p^2/2m$

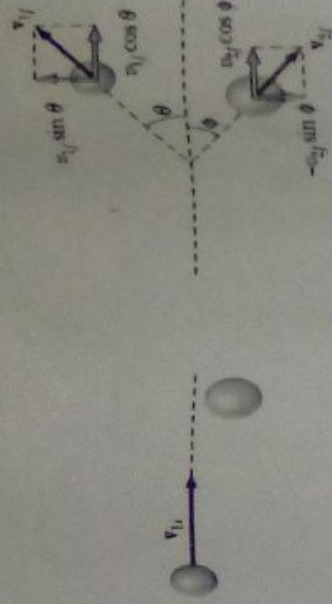
(b) Express the magnitude of the particle's momentum in terms of its kinetic energy and mass.

30. A 3.00-kg steel ball strikes a wall with a speed of 10.0 m/s at an angle of 60.0° with the surface. It bounces off with the same speed and angle (Fig. below). If the ball is in contact with the wall for 0.200 s, what is the average force exerted by the wall on the ball?



31. A proton collides elastically with another proton that is initially at rest. The incoming proton has an initial speed of 3.50×10^5 m/s and makes a glancing collision with the second proton, as in Figure below. (At close separations, the protons exert a repulsive electrostatic force on each other.) After the collision, one proton moves off at an angle of 37.0° to the original direction of motion, and the second deflects at an angle of $+$ to the same axis. Find the final speeds of the two protons and the angle $+$.

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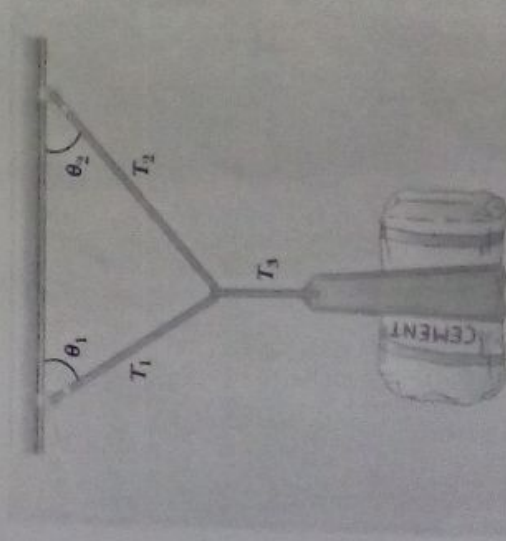


32. A 0.600-kg particle has a speed of 2.00 m/s at point "A" and kinetic energy of 7.50 J at point "B". What is
- Its kinetic energy at A?
 - Its speed at B?
 - The total work done on the particle as it moves from A to B?

33. A bag of cement of weight 325 N hangs from three wires as shown in Figure below.

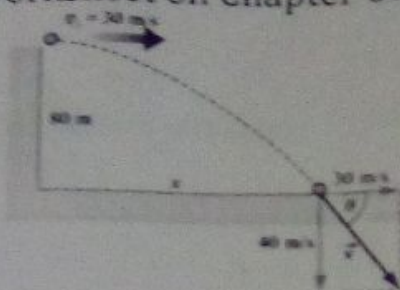
Two of the wires make angles

$\theta_1 = 60^\circ$ and $\theta_2 = 25^\circ$ with the horizontal. If the system is in equilibrium, find the tensions T_1 , T_2 , and T_3 in the wires.

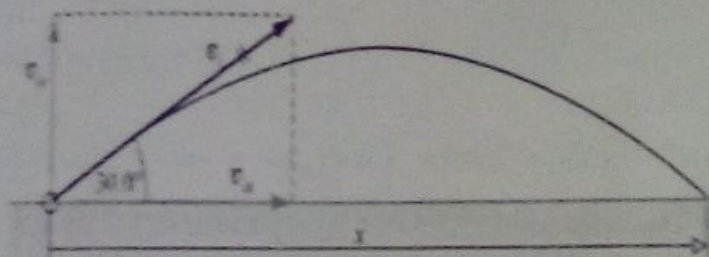


34. A 5.00-kg object placed on a frictionless, horizontal table is connected to a string that passes over a pulley and then is fastened to a hanging 9.00-kg object, as in Figure below. Draw free-body diagrams of both objects. Find the acceleration of the two objects and the tension in the string.

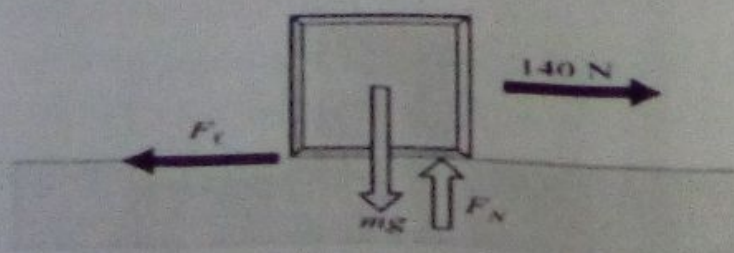
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10. A baseball is thrown with an initial velocity of 100 m/s at an angle of 30.0° above the horizontal. As shown in figure below. How far from the throwing point will the baseball attain its original level?



11. A 5.0 kg object is to be given an upward acceleration of 0.30 m/s^2 by a rope pulling straight upward on it. What must be the tension in the rope?
12. The only force acting on a 5 kg object has components $F_x = 20 \text{ N}$ and $F_y = 30 \text{ N}$. Find the acceleration of an object.
13. A 400 g block with an initial speed of 80 cm/s slides along a horizontal tabletop against a friction force of 0.70 N .
- How far will it slide before stopping?
 - What is the coefficient of friction between the block and the tabletop?
14. A 70 kg box is slid along the floor by a 400 N force as shown in figure below. The coefficient of friction between the box and the floor is 0.50 when the box is sliding. Find the acceleration of the box.



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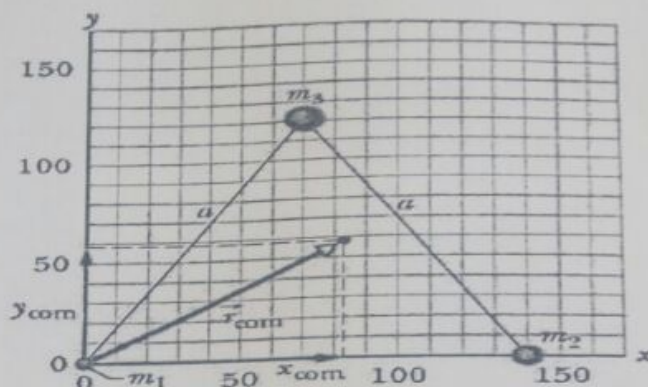
- Three forces that acts on a particle are given by $\vec{F}_1 = (20\hat{i} - 36\hat{j} + 73\hat{k})\text{N}$, $\vec{F}_2 = (-17\hat{i} + 21\hat{j} - 46\hat{k})\text{N}$ and $\vec{F}_3 = (-12\hat{k})\text{N}$. Find their resultant vector. Also find the magnitude of the resultant to two significant figures.
- If $\vec{A} = (-12\hat{i} + 25\hat{j} + 13\hat{k})$ and $\vec{B} = (-3\hat{i} + 7\hat{k})$, find the resultant when $\vec{A}$ is subtracted from $\vec{B}$.
- Change the speed 0.200 cm/s to units of kilometer per year?
- An object starts from rest with a constant acceleration of 8.00 m/s^2 along a straight line. Find
 - The speed at the end of 5.00 s .
 - The average speed for the 5 s interval
 - The distance traveled in the 5.00 s .
- A truck's speed increases uniformly from 15 km/hr to 60 km/hr in 20 s . determine
 - The average speed
 - The acceleration
 - The distance travel, all in units of meters and second
- A ball is dropped from rest at a height of 15m above the ground.
 - What is its speed before it hits the ground
 - How long does it take to reach the ground
- The speed of a train is reduced uniformly from 15 m/s to 7.0 m/s while traveling a distance of 90m .
 - Compute the acceleration
 - How much farther will the train travel before coming to rest, provided the acceleration remains constant?
- A baseball is thrown straight upward on the motion with an initial speed of 35 m/s . Compute
 - The maximum height reached by the ball
 - The time taken to reach that height
 - Its velocity 30 s after it is thrown
 - When the ball's height is 100m .
- As shown in figure blow a projectile is fired horizontally with a speed of 30 m/s from the top of a cliff 80m height.
 - How long will it take to strike the level ground at the base of the cliff?
 - How far from the foot of the cliff will it strike?
 - With what velocity will it strike?

17. A 2kg block initially at rest is pulled along a horizontal smooth surface by a constant force of 12N. What is
 a) the speed of the block after it travelled 3m? b) the acceleration of the block
 c) the time taken by the block to travel 3m

18. A car of mass 1500 kg initially moving at 15m/s collides with a wall and rebounds with a velocity of 3 m/s. If the collision lasts for 0.15 sec.

a) Find the impulse delivered to the car b) The average force exerted on the car

19. Three particles of masses $m_1 = 1.2$ kg, $m_2 = 2.5$ kg, and $m_3 = 3.4$ kg form an equilateral triangle of edge length $a = 140$ cm. Where is the center of mass of this system, as shown in the figure below?)



20. Given that Moon orbit earth each 27.3 day and that it is an average distance of 3.84×10^8 m from the center of the earth. What is the period of an artificial satellite orbiting an average altitude of 1500km above the earth's surface?

21. A large man and small boy stand facing each other on frictionless ice. They put their hands together and push against each other, so that they move apart. Who moves away the higher speed? Why?

22. When an object is immersed in a liquid at rest, why is the net force on the object in the horizontal direction equal to zero?

23. Explain strain, stress and elastic modulus.

24. If the shear stress in steel exceeds 4×10^8 N/m², the steel ruptures, Determine the shearing force necessary to, a) shear steel bolt 1cm in diameter and b) punch a 1cm diameter hole in a steel plate 0.5cm thick.

25. A 200kg load is hung on a wire having length 4m, cross-sectional area of 0.2×10^{-4} m² and the Young's Modulus 8×10^{10} N/m². What is its increase in length?

26. In a hydraulic press the small piston has an area of 5cm² while the larger piston has an area of 200 cm². If a force of 80 N is applied on the small piston, what load can it lift up?

27. A bullet of mass 0.2kg is fired horizontally at 120m/s, in to a stationary block of wood that is free to move on a smooth horizontal plane. The wooden block with the bullet embedded in it, moves off with the speed of 10m/s, find the mass of the block?

28. A ball of mass 0.1kg has a velocity of 120m/s; a ball of mass 0.2 kg has a velocity of 0.6m/s. They meet in a head on elastic collision. (a) Find their velocities after the collision. (b) Find the velocity of their center of mass before and after the collision.

29. At a certain point in a pipe line the velocity is 2m/s at a height of 30m and the gauge pressure is 4×10^5 N/m², what will be the gauge pressure at the second point in the line 10m lower than the first if, the cross-sectional area at the second point is quarter that at the first, if the liquid in the pipe is water.

9. If vector $\vec{B}$ is added to vector $\vec{A}$, which two of the following choices must be true in order for the resultant vector to be equal to zero? (a) $\vec{A}$ and $\vec{B}$ are parallel and in the same direction. (b) $\vec{A}$ and $\vec{B}$ are parallel and in opposite directions. (c) $\vec{A}$ and $\vec{B}$ have the same magnitude. (d) $\vec{A}$ and $\vec{B}$ are perpendicular.
10. Suppose you are running at constant velocity and you wish to throw a ball such that you will catch it as it comes back down. In what direction should you throw the ball relative to the ground?
11. In an inertial frame of reference, if an object doesn't have an acceleration then which of the following can be true about the condition of the object?
 - a. Unbalanced force acts on the object
 - b. The object is moving with an arbitrary velocity
 - c. Forces act on the object, but the forces cancel out
12. Which of the following is the reaction force to the gravitational force acting on your body as you sit in your chair?
 - a. The normal force exerted by the chair
 - b. The force you exert downward on the seat of the chair
 - c. Neither of these forces
13. If vector $\mathbf{A}$ equals vector $\mathbf{B}$, what can you conclude about the components of $\mathbf{A}$ and $\mathbf{B}$?
14. If $\mathbf{A} \times \mathbf{B} = 8\mathbf{i} - 14\mathbf{j} + \mathbf{k}$ and $\mathbf{A} + \mathbf{B} = 5\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$, find the vectors $\mathbf{A}$ and $\mathbf{B}$.
15. If you push a heavy box which is at rest, you must exert some more force to start its motion. However, once the box is sliding, you can apply a smaller force to maintain that motion. Why?
16. State and explain the work-energy theorem.
17. If a bird collides with the windshield of a fast-moving bus, which one experiences an impact force with a larger magnitude? Which one experienced the greater acceleration?
18. Can the torque on a particle be zero without the force being zero? Explain.
19. Can the force on a particle be zero without the angular momentum being zero? Explain.
20. What is the difference between shear and stress?
21. When a smooth-flowing stream of water comes out of a faucet, it narrows as it falls. Why does it do so?
22. A very smooth wooden block is pressed against the bottom of a container of water so that no water remains under the block. Will this block rise to the surface when it is released? Why? (Hint: can the water exert a force on the bottom of the block?)
23. If the absolute pressure at a depth d in a lake is P , the absolute pressure at a depth $2d$ will be _____. What if there were a gauge pressure instead of absolute pressure?

Part II: Work Out Problems

1. Find the sum and difference of two position vectors $\vec{A}$ and $\vec{B}$ in space and given by $\vec{A} = (2\mathbf{i} + 4\mathbf{j})\text{m}$; and $\vec{B} = (3\mathbf{i} + 4\mathbf{j} + 4\mathbf{k})\text{m}$.

2. If the polar coordinates of the point (x, y) are (r, θ) , determine the polar coordinates for the points: $(-x, y)$, $(-2x, -2y)$ and $(3x, -3y)$.
3. A particle starts from the origin at $t=0$ with an initial velocity having an x-component of 15 m/s and a y-component of -15 m/s. The particle moves in the xy plane with an x-component of acceleration only, given by $a_x = 4.0 \text{ m/s}^2$.
 - a. Determine the components of the velocity vector at any time and the total velocity vector at any time.
 - b. Calculate the velocity and speed of the particle at $t=5.0 \text{ s}$.
4. A hiker begins a trip by first walking 20.0 km southeast from her car. She stops and sets up her tent for the night. On the second day, she walks 30.0 km in a direction 60.0° north of east, at which point she discovers a forest ranger's tower.
 - a. Determine the components of the hiker's displacement for each day
 - b. Determine the components of the hiker's resultant displacement $\mathbf{R}$ for the trip. Find an expression for $\mathbf{R}$ in terms of unit vectors.
5. Show that the trajectory of a projectile motion is parabolic.
6. A plane drops a package of supplies to a party of explorer on the ground. If the plane is traveling horizontally at 40.0 m/s and is 100 m above the ground, where does the package strike the ground relative to the point at which it is released?
7. A projectile is fired at an angle of 30° from the horizontal with some initial speed. Firing the projectile at what other angle results in the same horizontal range if the initial speed is the same in both cases? Neglect air resistance.
8. A projectile is thrown from the top of a building upwards at an angle of 60° to the horizontal with initial speed of 20.0 m/s. The height from which the stone is thrown is 45.0 m above the ground.
 - a. How long does it take the stone to reach the ground?
 - b. What is the speed of the stone just before it strikes the ground?
9. Two block of masses m_1 and m_2 , with $m_1 > m_2$, are placed in contact with each other on a frictionless, horizontal surface. A constant horizontal force F is applied to m_1 .
 - a. Find the magnitude of the acceleration of the system.
 - b. Determine the magnitude of the contact force between the two blocks.
10. Find the constant force in the (a) cgs unit system and (b) SI unit system needed to accelerate a mass of 10, 000 g moving along a straight line from a speed of 54 km/hr to 108 km/h in 5 minutes.
11. Find the work done by the force in the above problem (problem 10).
12. A mass of 5000 kg moves in a straight line from a speed of 540 km/hr to 720 km/hr in 2 minutes. What is the impulse developed in this time?
13. What is the magnitude of the impulse developed by a mass of 200 g which changes its velocity from $5\mathbf{i} - 3\mathbf{j} + 7\mathbf{k}$ m/s to $2\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ m/s?
14. A particle is acted upon by the forces $\mathbf{F}_1 = 5\mathbf{i} - 10\mathbf{j} + 15\mathbf{k}$, $\mathbf{F}_2 = 10\mathbf{i} - 25\mathbf{j} - 20\mathbf{k}$ and $\mathbf{F}_3 = 15\mathbf{i} - 20\mathbf{j} + 10\mathbf{k}$. Find the forces needed to keep the particle at equilibrium.

MIZAN-TEPI UNIVERSITY COLLEGE OF NATURAL & COMPUTATIONAL SCIENCE
Department of Physics General Physics (Phys1011) Work-Sheet for First Year Regular Natural
Science Students (2012E.C First Semester Academic Year)

1. On an interstate highway in a rural region of Mizan-Tepi, a car is traveling at a speed of 38.0 m/s. Is the car exceeding the speed limit of 75.0 mi/h?
2. Give a dimensional expression for Energy, linear momentum and power
3. How many significant figures do the following numbers have?
 a) 250.08 b) 0.200 c) 12000 d) 0.08043 e) 45632
4. Eyob and John have measured the voltage across a resistor to be 5.26 V and the current flowing through it to be 0.41 A. They work out the resistance ($R = V/I$). Eyob says that the resistance is 12.8 Ω . John disagrees and says that the resistance is 13 Ω . Who is correct? Explain your answer.
5. In what type of a motion the instantaneous velocity and average velocity are the same, and instantaneous acceleration and average acceleration are the same?
6. Given two vectors **A** and **B** lying in the x-y plane by $A = 5i + 3j$ and $B = -2i + j$, what is the magnitude of $A+B$, and a unit vector in the direction of the resultant vector
7. If $A = 2i - 3j$, $B = -3i + 3j$ and $C = 6i + 9j$, what are the values of **a** and **b** respectively, so that $aA + bB + C = 0$
8. Given two vectors $A = 3i + 2j - 5k$ and $B = i + j + 2k$. What is the magnitude of $A+2B$ or $|A+2B|$?
9. Vectors **A** = 10 unit due East and **B** = 8 unit make 60° above the positive x-axis. What is the magnitude and direction of the resultant vector?
10. What should be the angle b/n two non-zero vectors **A** and **B** of equal magnitude for
 i) The magnitude of their sum to be equal to **|A|**
 ii) The magnitude of their sum to be equal to **2|A|**
11. A bicyclist encounters a series of hills. Uphill with constant speed of V_1 and with down ward constant speed of " V_2 ". The total distance he travelled is "**X**", with uphill and downhill portions of equal length. Show that the cyclist average speed is $\frac{2(V_1 \cdot V_2)}{V_1 + V_2}$
12. A particle moves along the **X-axis** in such a way that its distance as a function of time is given by $X(t) = t^2 - 12t + 6$. Where X- is displacement in meter and t-is time in second.
 Determine a) the average velocity b/n $t=0$ sec and $t = 2$ sec b) the velocity of a particle at $t = 3$ second c) the average acceleration b/n $t = 0$ sec, and $t=2$ sec. d) the acceleration of the particle at $t = 3$ second e) by comparing your result on part C and D, what do you say about the motion of the particle?
13. An aero plane taxis onto the runway going at 10m/s. If it can accelerate steadily at $3.0m/s^2$ and its take-off speed 90m/s, what length of runway will it need?
14. A Speedy tortoise can run with a velocity of 10cm/s, and a hare can run at 2m/s. In a race they both start at the same time, but the hare stops for 2min. The tortoise wins by its neck. What is the length of the race?
15. A projectile is fired in such a way that its horizontal range is equal to four times its maximum height. What is angle of projection?
16. Discuss the following a) The three keplers' Law of planetary motion b) kinetic and static friction
 c) Weightlessness

Jimma University
College of Natural Sciences
Physics Department

**Worksheet I on the Course "General Physics" for Natural Sciences
Freshman Program**

Academic year: 2019/20

Part I: Conceptual Questions

1. Under which of the following conditions is the magnitude of the average velocity of a particle moving in one dimension smaller than the average speed over some time interval?
 - a) A particle moves in the $+x$ direction without reversing
 - b) A particle moves in $-x$ direction without reversing
 - c) A particle moves in the $+x$ direction and then reverses the direction of its motion
 - d) There are no conditions for which this is true
2. Consider the following one-dimensional motions: (A) a ball thrown directly upward rises to a highest point and falls back into the thrower's hand; (B) a race car starts from rest and speeds up to 100 m/s ; (C) a spacecraft drifts through space at constant velocity. Are there any points in the motions of these objects at which the instantaneous velocity has the same value as the average velocity over the entire motion? If so, identify the point(s).
3. If a car is travelling eastward and slowing down, what is the direction of the force on the car that causes it to slow down? (a) Eastward (b) Westward (c) neither of these.
4. Which of the following is true? (a) if car is travelling eastward, its acceleration is eastward, (b) if a car is slowing down, its acceleration must be negative, (c) a particle with constant acceleration can never stop and stay stopped.
5. If the average velocity of an object is zero in some time interval, what can you say about the displacement of the object for that time interval?
6. Two cars are moving in the same direction in parallel lanes along a highway. At some instant, the velocity of car A exceeds the velocity of car B. Does this mean that the acceleration of A is greater than that of B? Explain.
7. A student at the top of a building of height h throws on ball upward and with a speed of v_i and then throws a second ball downward with the same initial speed, v_i . How do the final velocities of the balls compare when they reach the ground?
8. If vector $\vec{B}$ is added to vector $\vec{A}$, under what condition does the resultant vector $\vec{A} + \vec{B}$ have magnitude $A+B$? (a) $\vec{A}$ and $\vec{B}$ are parallel and in the same direction. (b) $\vec{A}$ and $\vec{B}$ are parallel and in opposite directions. (c) $\vec{A}$ and $\vec{B}$ are perpendicular.